

Supplementary Information for Information-Cost Diagnostics for Threshold Instability in Probabilistic Classifiers

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Overview

This supplementary file contains the detailed figures generated by the script “generate_multidomain_stability_figures.py”. The main manuscript reports the all-threshold heatmaps and focuses the detailed progressive analysis on $\tau = 0.5$. This supplement provides the threshold-specific diagnostic scatter plots for $\tau = 0.3$, $\tau = 0.5$ and $\tau = 0.7$, together with final ROC curves for each dataset-model pair.

The threshold-certificate scatter plots include flipped points only. The horizontal axis is the conservative Pinsker certificate,

$$IG_{\min} = 2|p_{\text{prev}} - \tau|^2,$$

and the vertical axis is the observed Bernoulli KL displacement,

$$IG_{\text{step}} = D_{\text{KL}}(p_{\text{curr}} \| (p_{\text{prev}})).$$

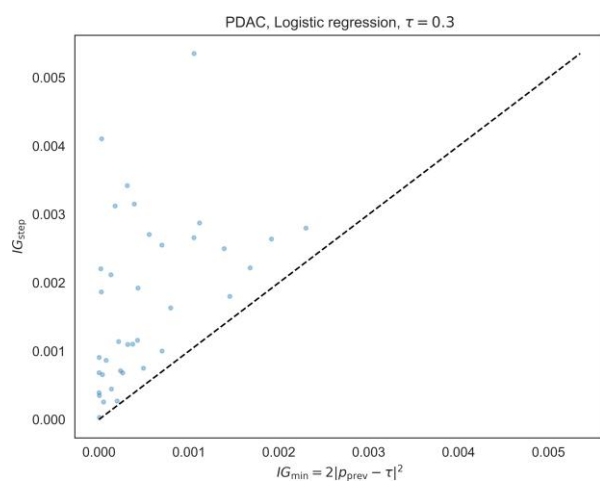
The dashed diagonal corresponds to $IG_{\text{step}} = IG_{\min}$. Boundary satisfaction is deterministic for threshold-crossing Bernoulli probabilities; the scatter plots show how far the observed flips lie above the conservative certificate.

Table S1: Dataset-model-threshold combinations for which no threshold-certificate scatter figure is expected because no flipped points were observed in the reported run.

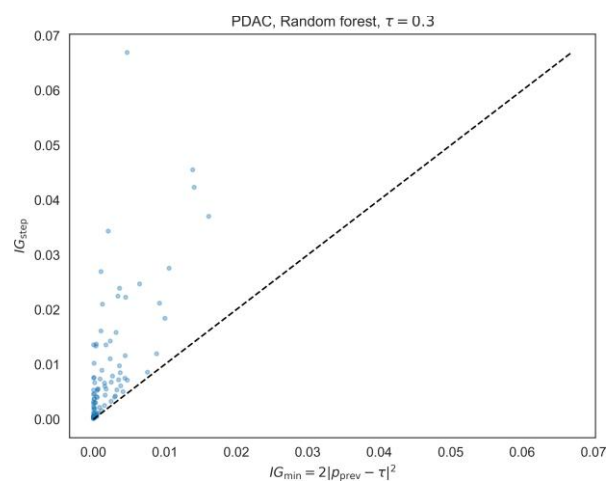
Dataset	Model	Threshold
Machine failure	Logistic regression	$\tau = 0.5$
Machine failure	Logistic regression	$\tau = 0.7$
Machine failure	ANN	$\tau = 0.7$

1 Threshold-certificate scatter plots

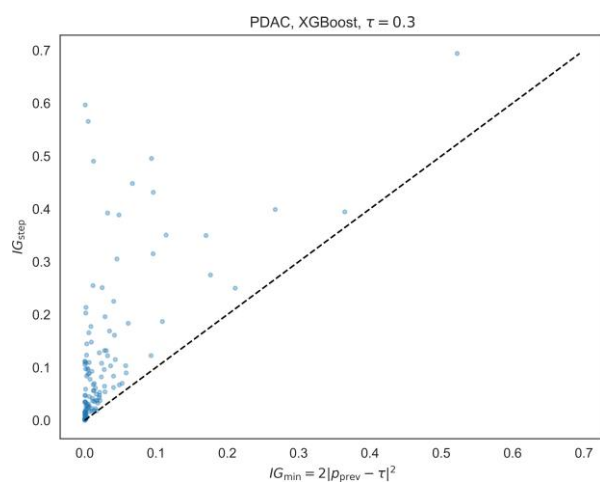
1.1 PDAC biomarkers



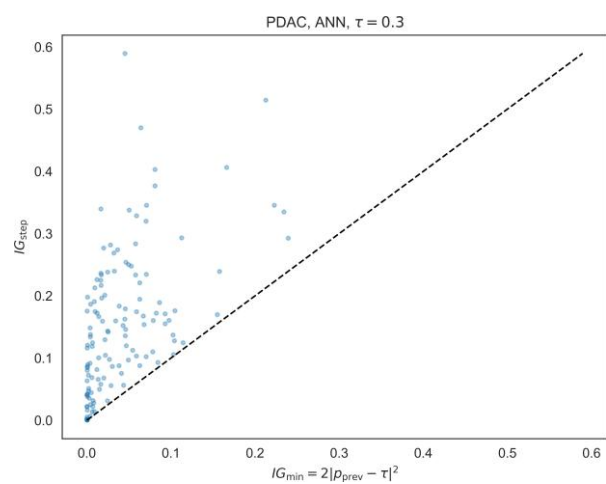
(a) Logistic regression



(b) Random forest

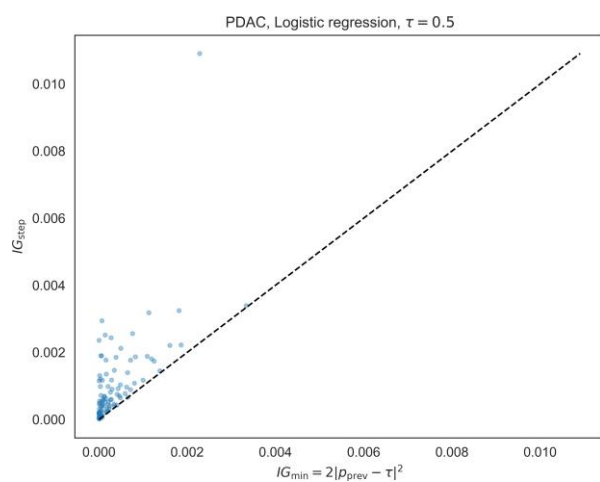


(c) XGBoost

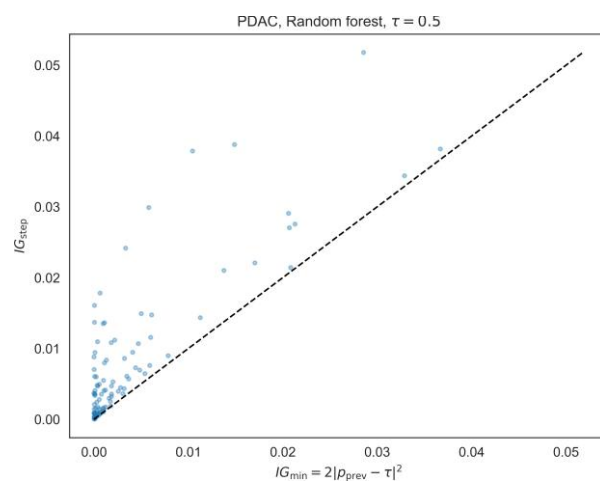


(d) ANN

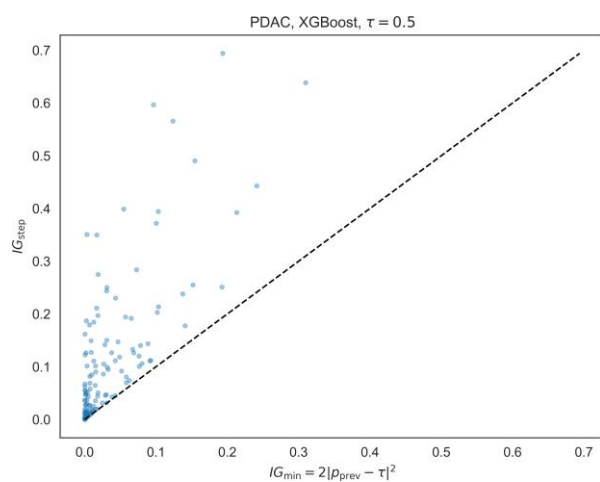
Figure S1: Threshold-certificate scatter diagnostics for the PDAC biomarker dataset at $\tau = 0.3$.



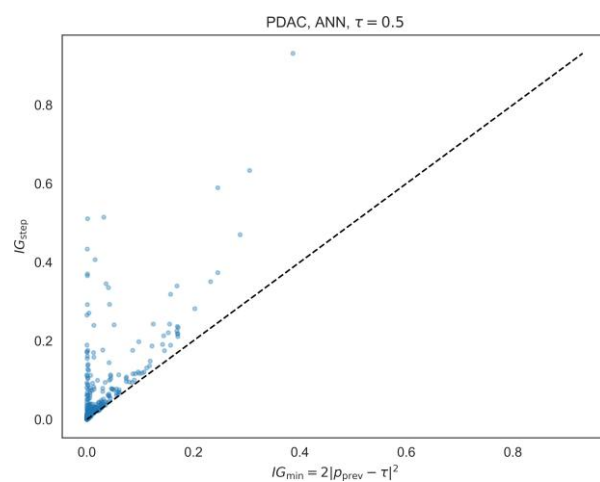
(a) Logistic regression



(b) Random forest

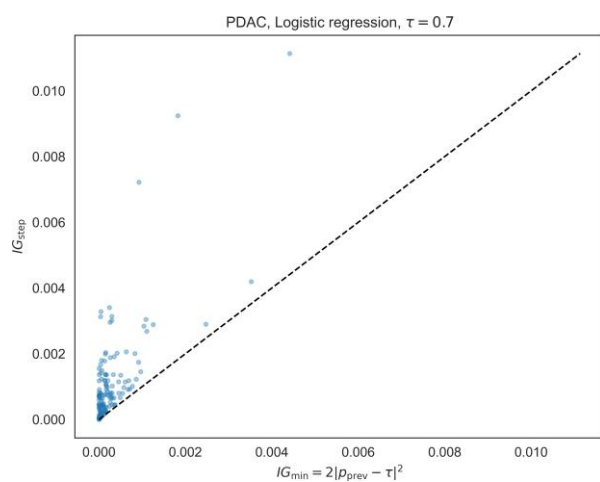


(c) XGBoost

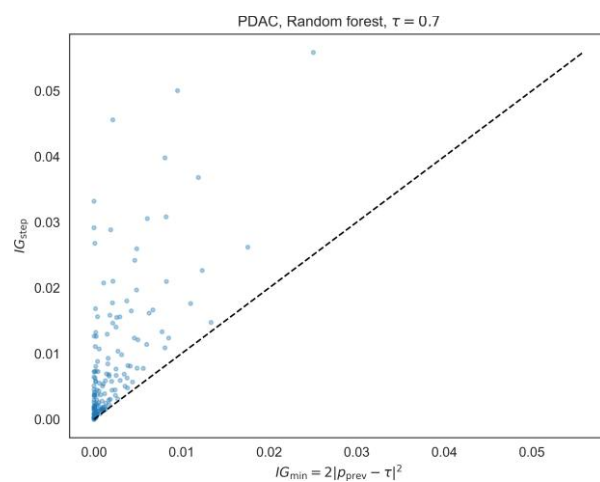


(d) ANN

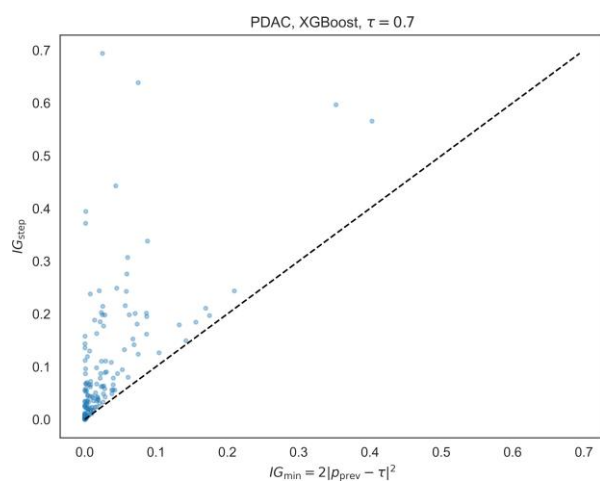
Figure S2: Threshold-certificate scatter diagnostics for the PDAC biomarker dataset at $\tau = 0.5$.



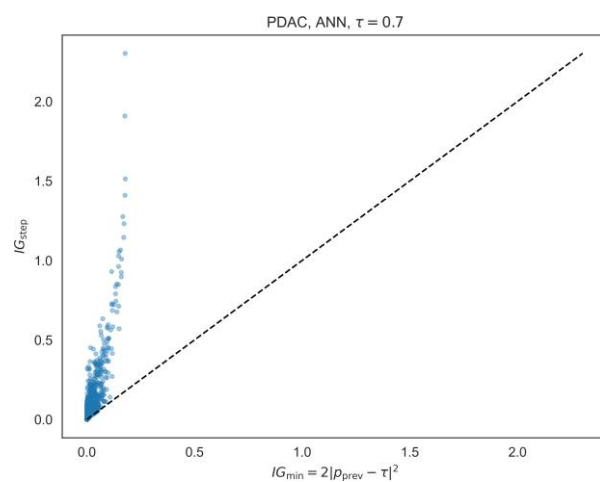
(a) Logistic regression



(b) Random forest



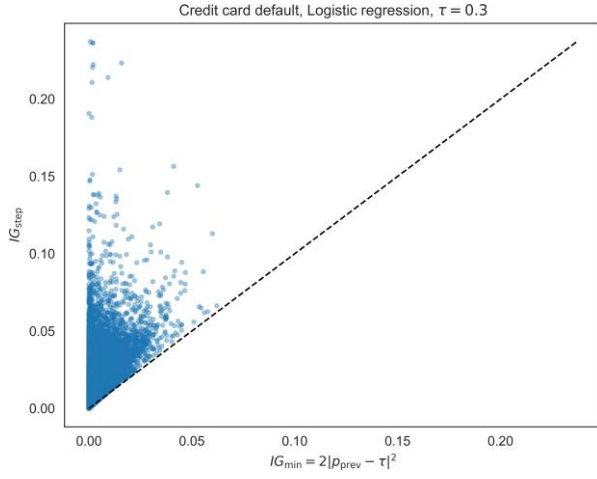
(c) XGBoost



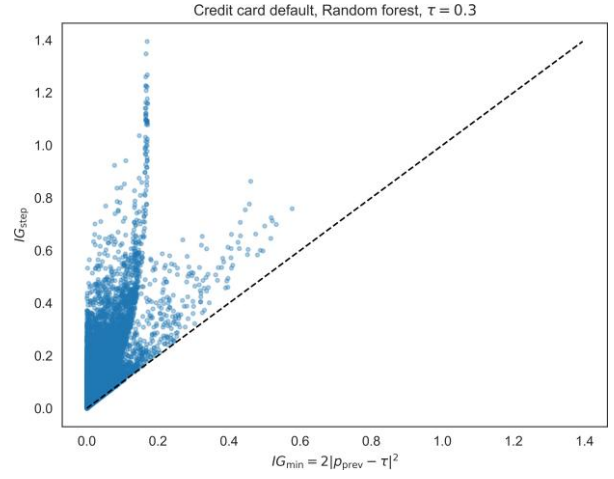
(d) ANN

Figure S3: Threshold-certificate scatter diagnostics for the PDAC biomarker dataset at $\tau = 0.7$.

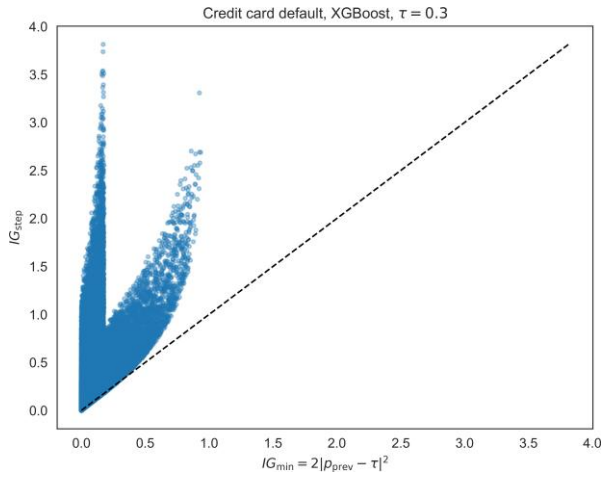
1.2 Credit card default



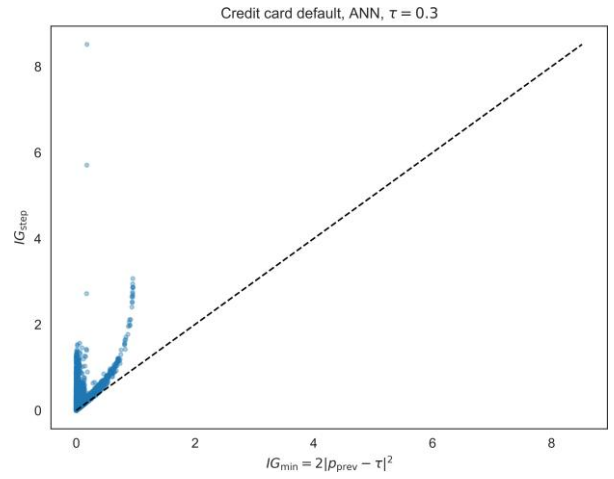
(a) Logistic regression



(b) Random forest

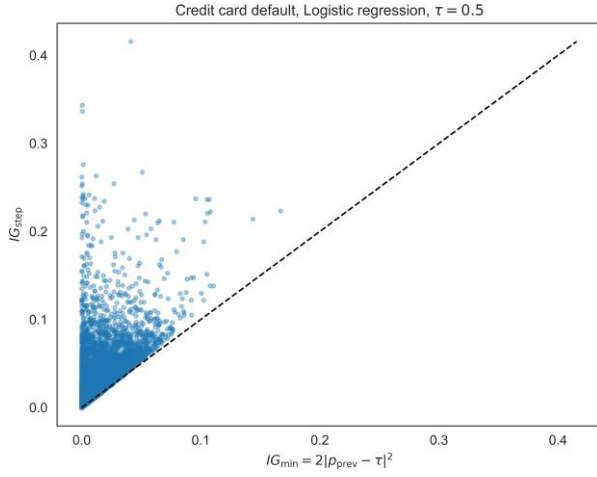


(c) XGBoost

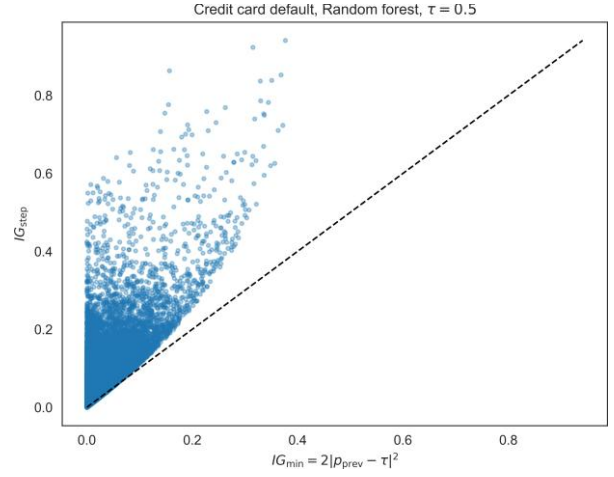


(d) ANN

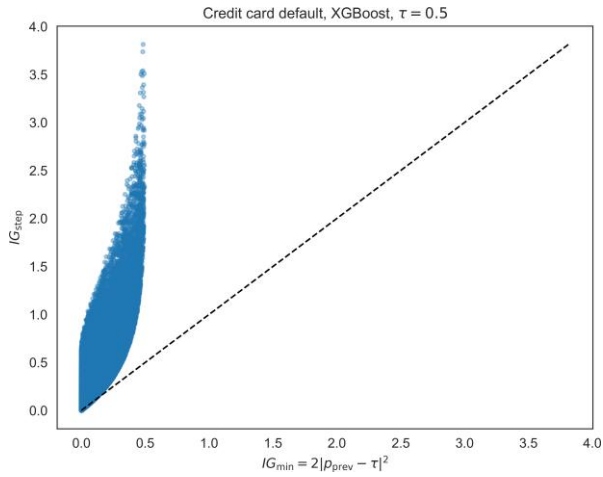
Figure S4: Threshold-certificate scatter diagnostics for the credit-card-default dataset at $\tau = 0.3$.



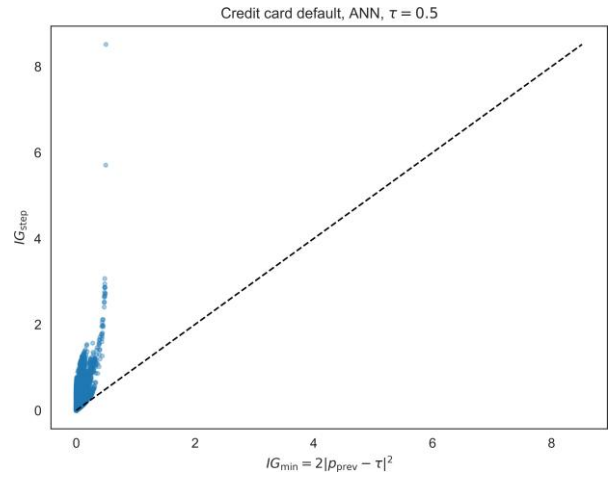
(a) Logistic regression



(b) Random forest

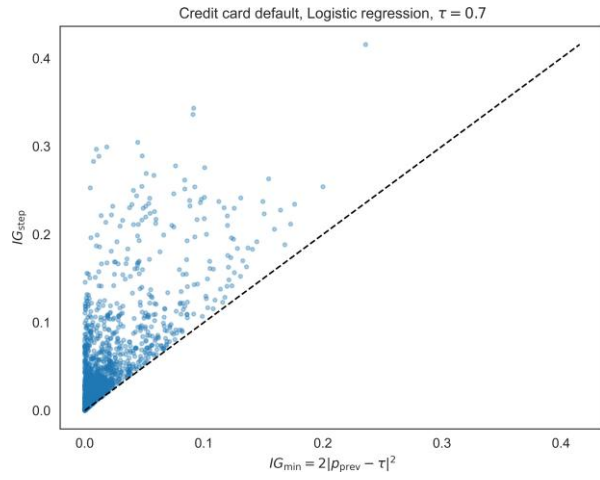


(c) XGBoost

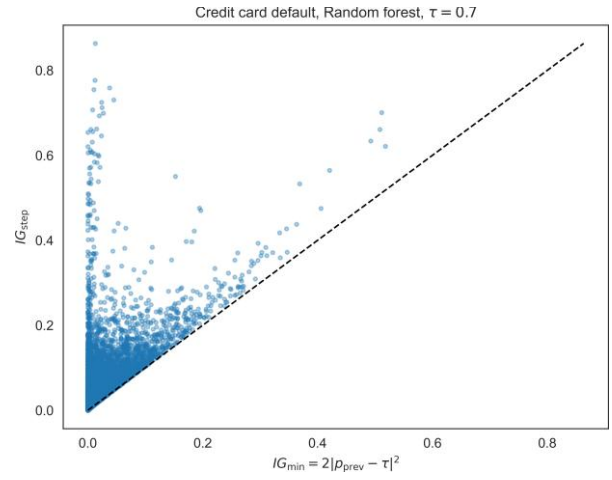


(d) ANN

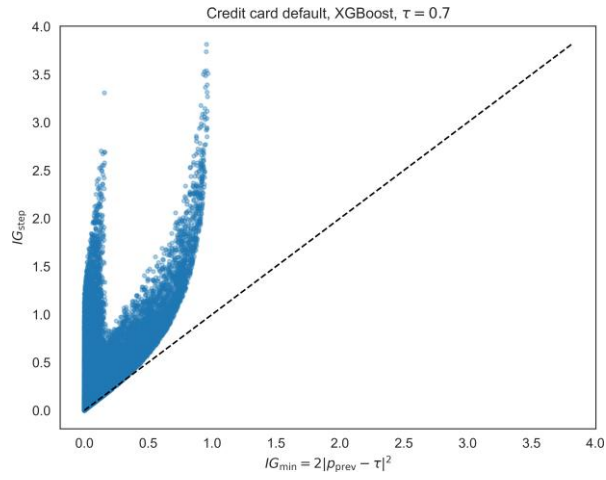
Figure S5: Threshold-certificate scatter diagnostics for the credit-card-default dataset at $\tau = 0.5$.



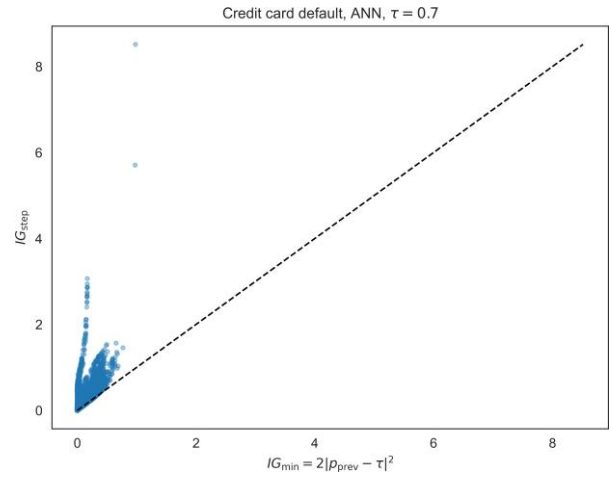
(a) Logistic regression



(b) Random forest



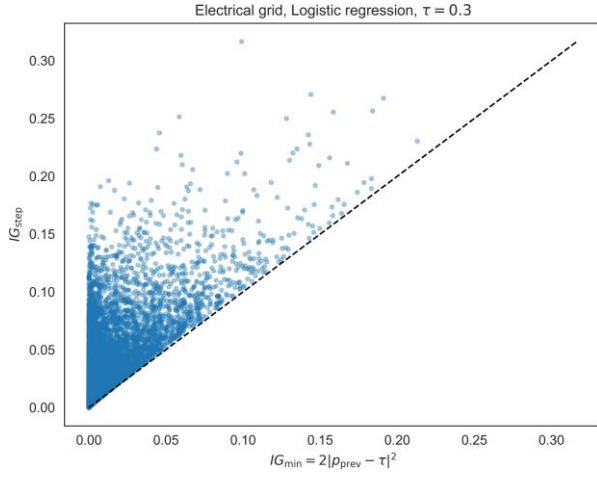
(c) XGBoost



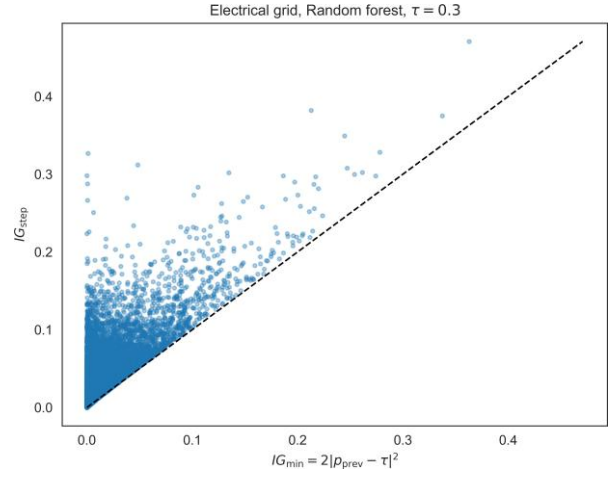
(d) ANN

Figure S6: Threshold-certificate scatter diagnostics for the credit-card-default dataset at $\tau = 0.7$.

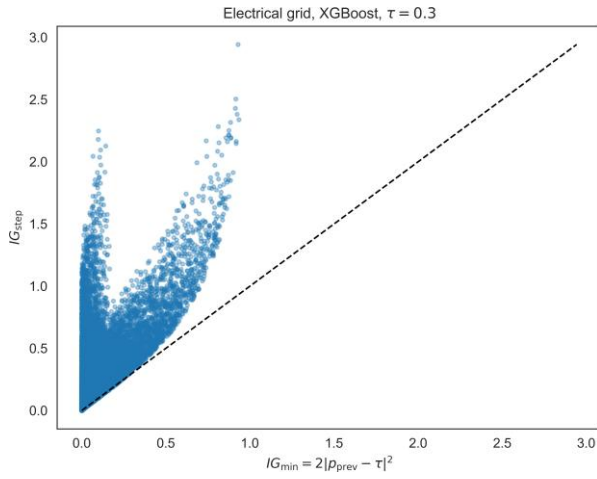
1.3 Electrical grid stability



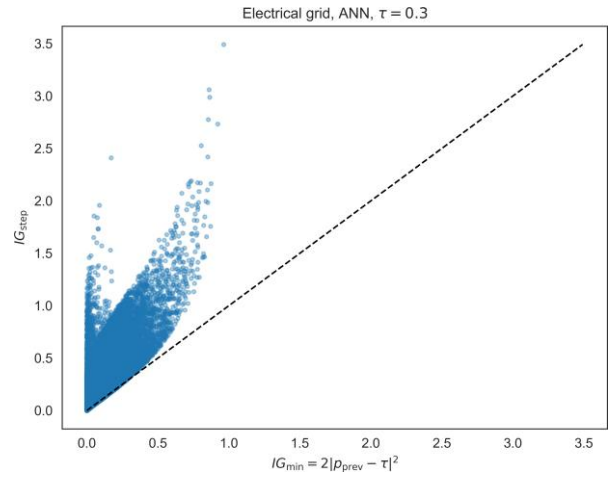
(a) Logistic regression



(b) Random forest

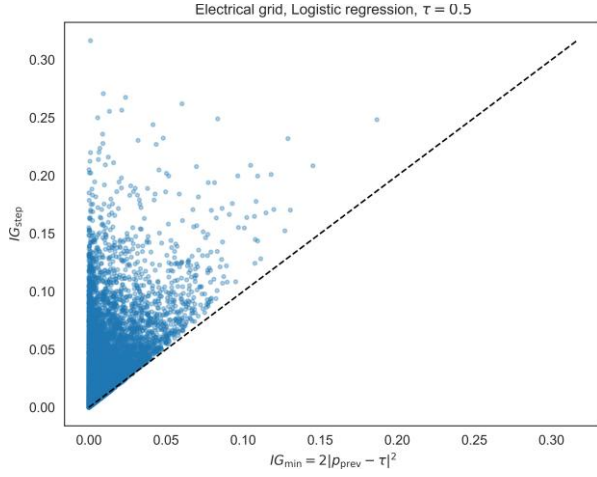


(c) XGBoost

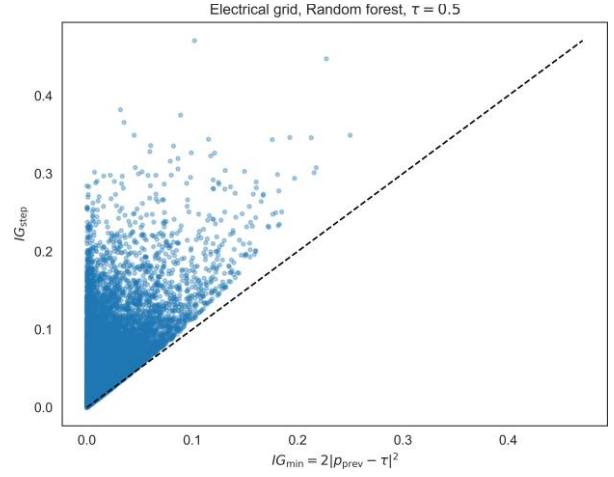


(d) ANN

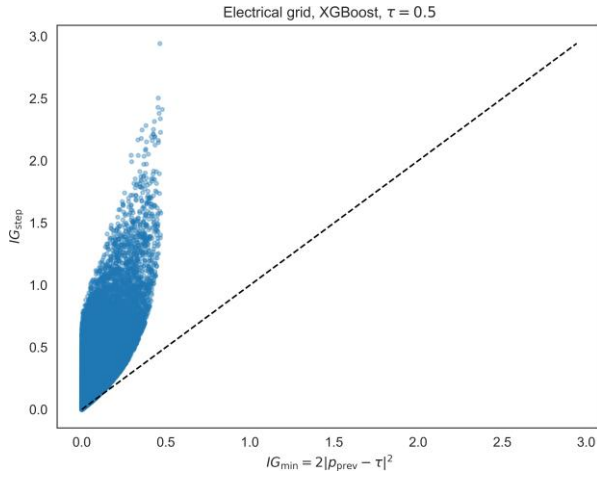
Figure S7: Threshold-certificate scatter diagnostics for the electrical-grid-stability dataset at $\tau = 0.3$.



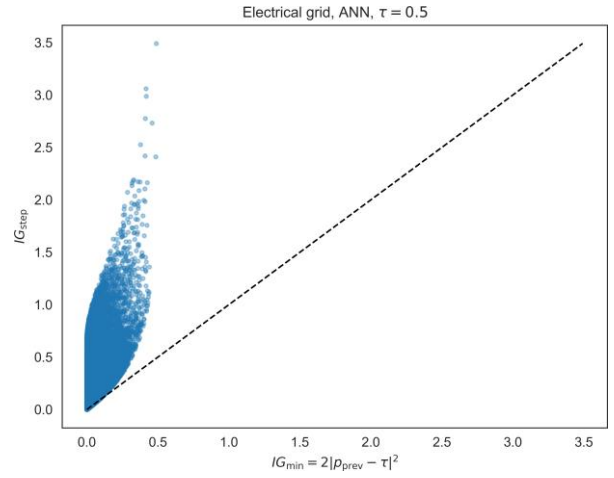
(a) Logistic regression



(b) Random forest

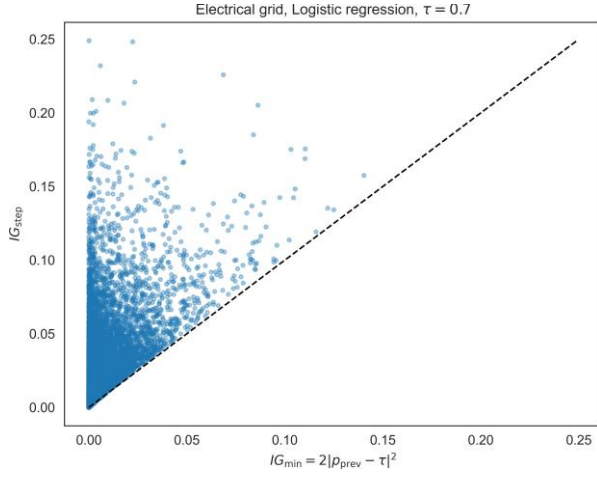


(c) XGBoost

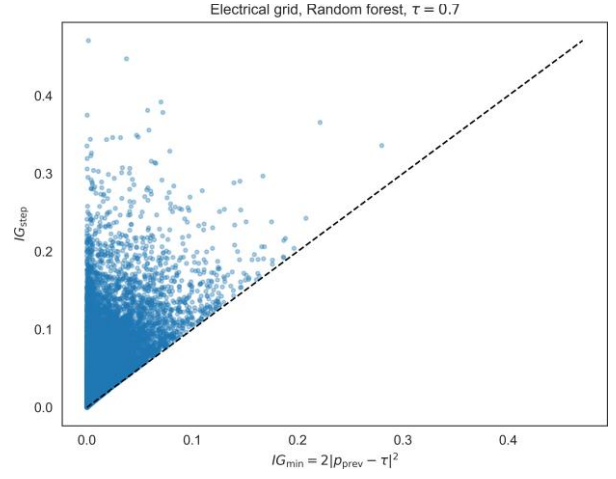


(d) ANN

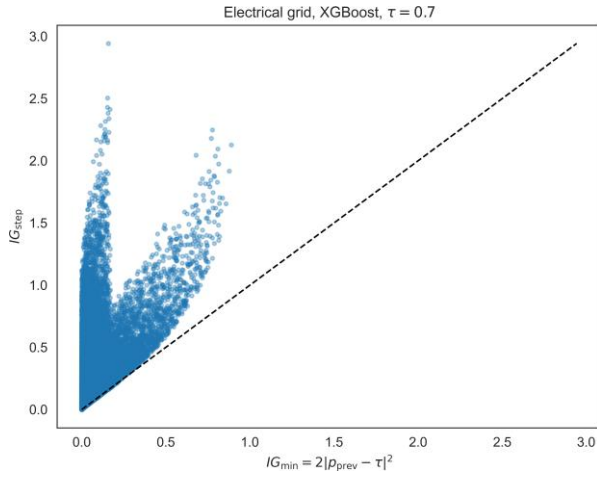
Figure S8: Threshold-certificate scatter diagnostics for the electrical-grid-stability dataset at $\tau = 0.5$.



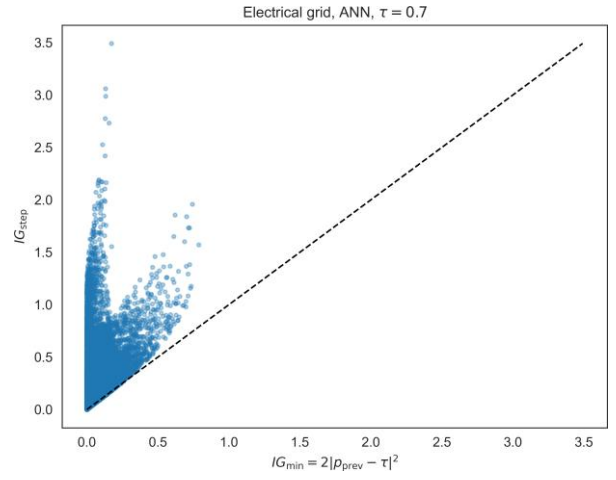
(a) Logistic regression



(b) Random forest



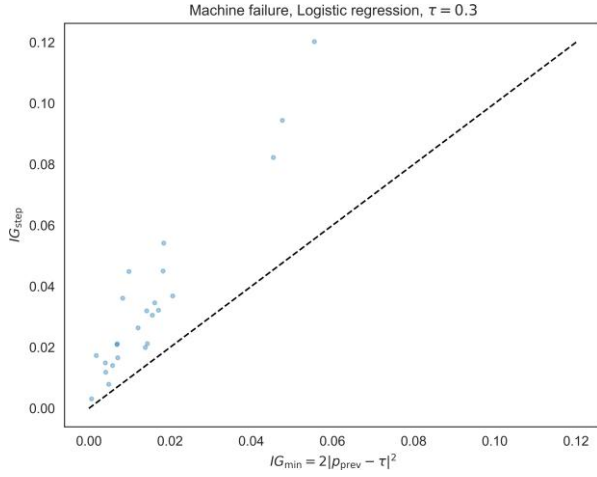
(c) XGBoost



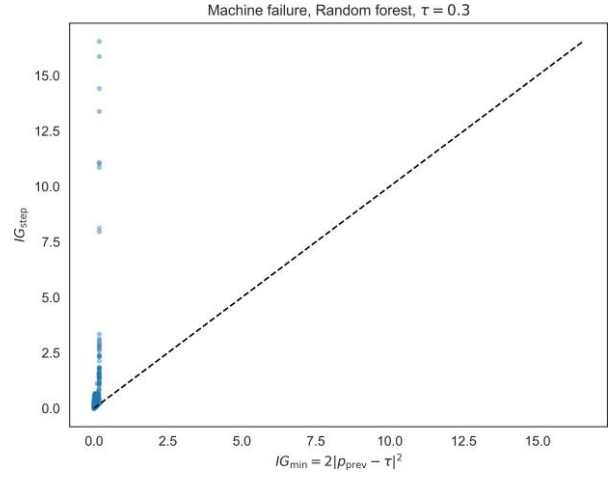
(d) ANN

Figure S9: Threshold-certificate scatter diagnostics for the electrical-grid-stability dataset at $\tau = 0.7$.

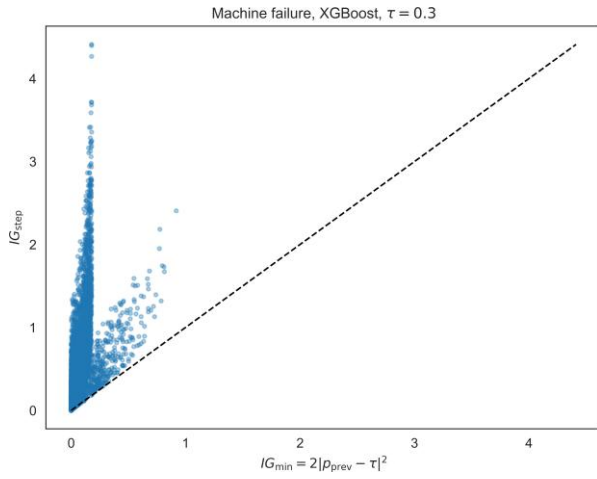
1.4 Machine failure



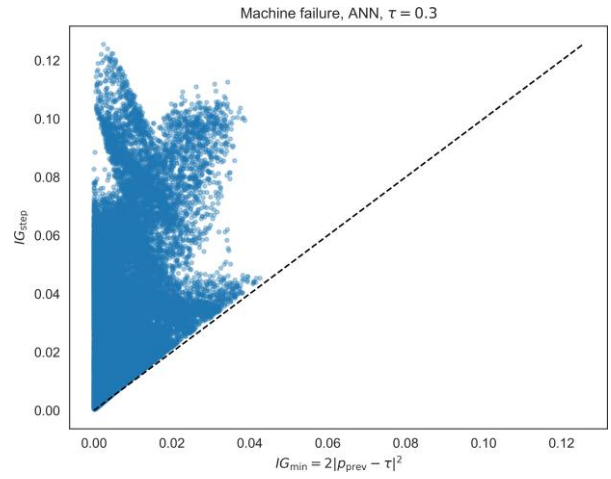
(a) Logistic regression



(b) Random forest



(c) XGBoost



(d) ANN

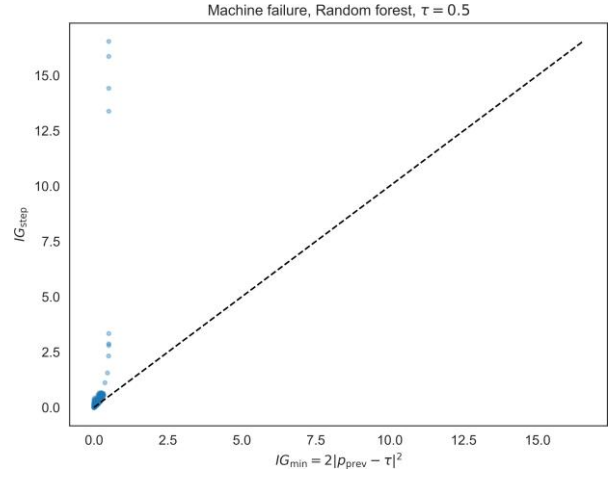
Figure S10: Threshold-certificate scatter diagnostics for the machine-failure dataset at $\tau = 0.3$.

No flipped points were observed for

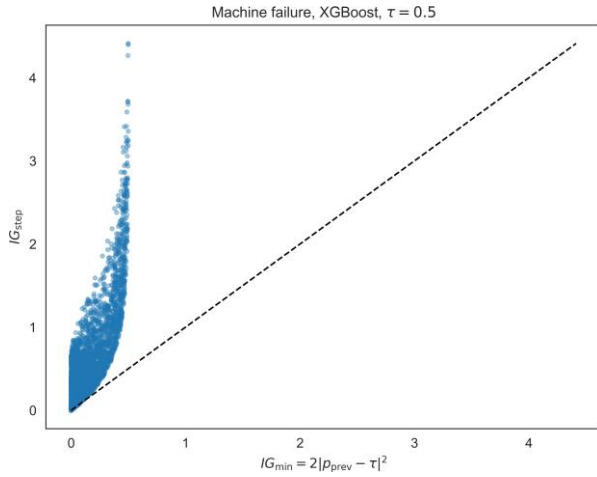
Logistic regression.

The Python script therefore did not generate a threshold-certificate scatter plot for this combination.

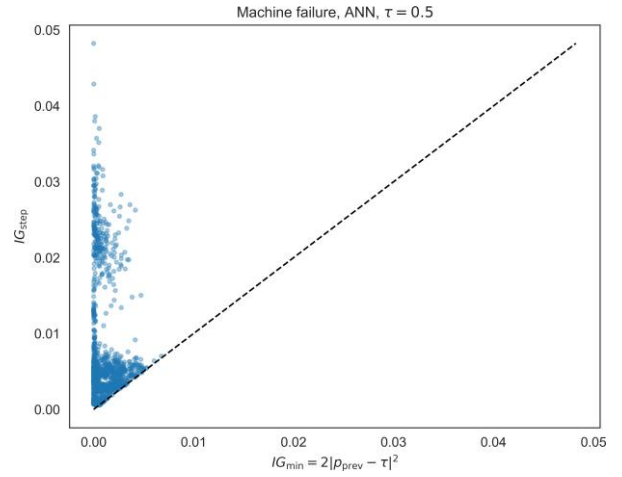
(a) Logistic regression



(b) Random forest



(c) XGBoost

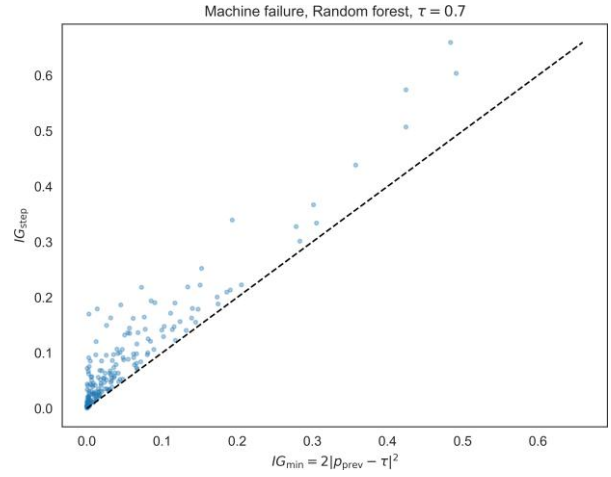


(d) ANN

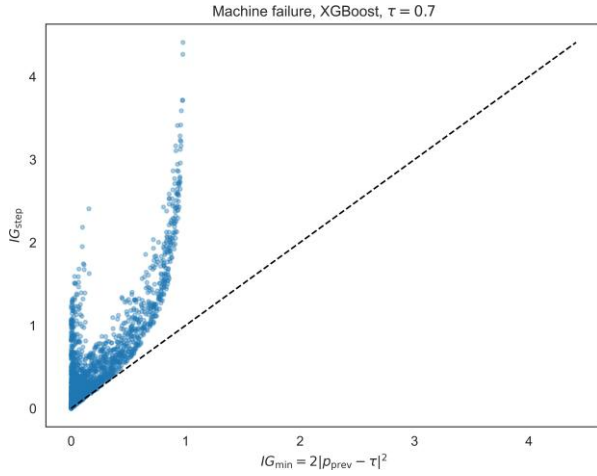
Figure S11: Threshold-certificate scatter diagnostics for the machine-failure dataset at $\tau = 0.5$. Logistic regression produced no flipped points in this setting, so no scatter plot was generated for that panel.

No flipped points were observed for
Logistic regression.
 The Python script therefore did not generate a
 threshold-certificate scatter plot for this combination.

(a) Logistic regression



(b) Random forest



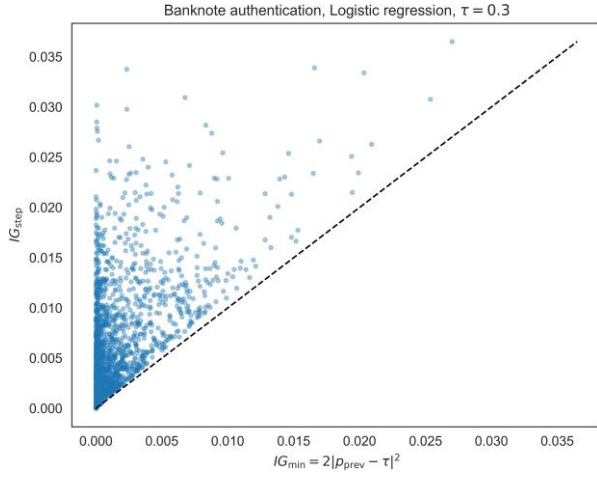
(c) XGBoost

No flipped points were observed for
ANN.
 The Python script therefore did not generate a
 threshold-certificate scatter plot for this combination.

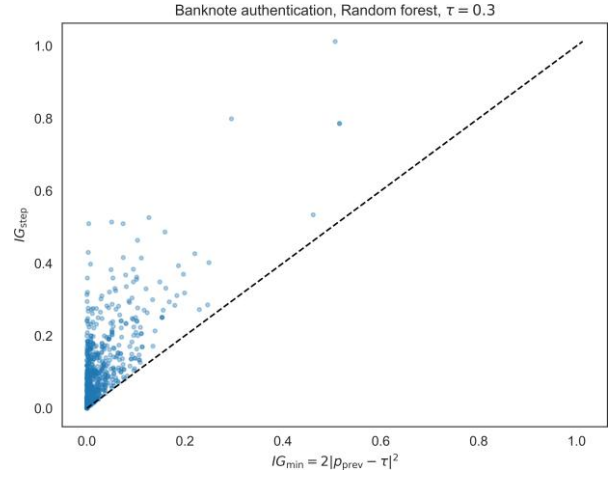
(d) ANN

Figure S12: Threshold-certificate scatter diagnostics for the machine-failure dataset at $\tau = 0.7$. Logistic regression and ANN produced no flipped points in this setting, so no scatter plots were generated for those panels.

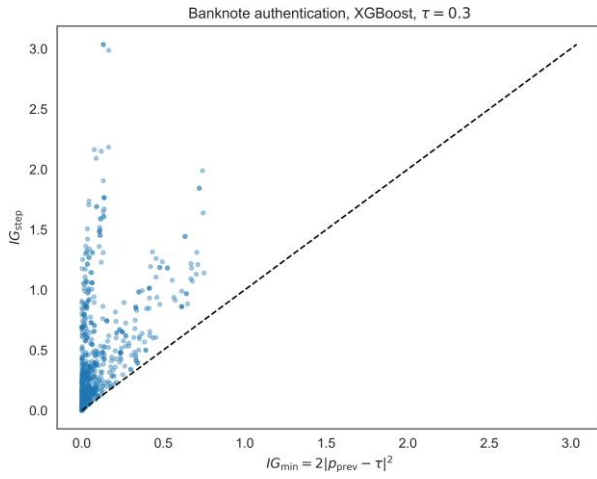
1.5 Banknote authentication



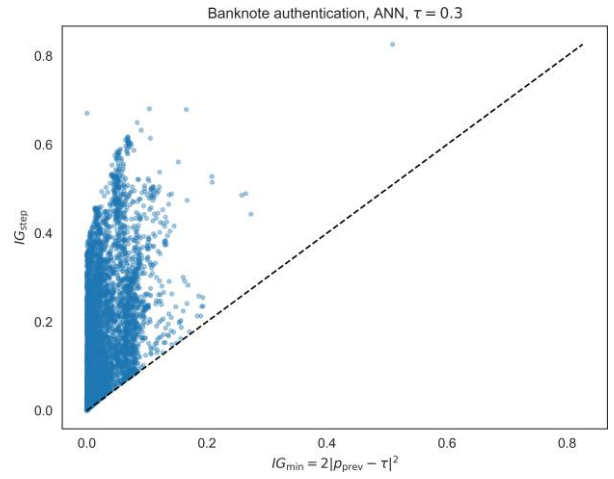
(a) Logistic regression



(b) Random forest

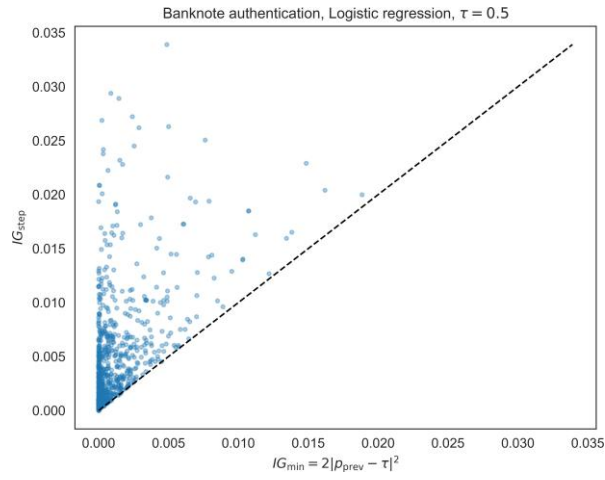


(c) XGBoost

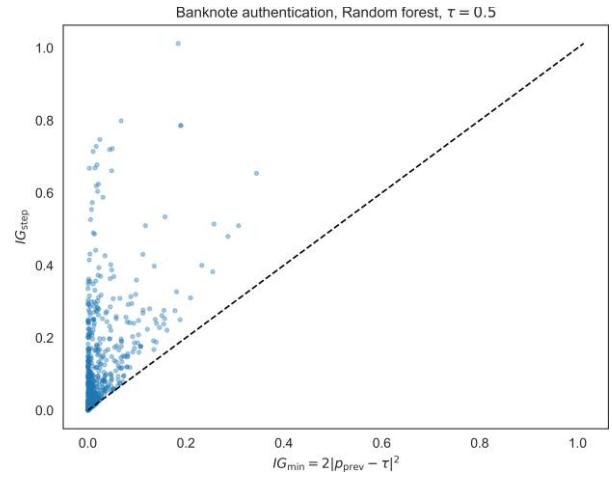


(d) ANN

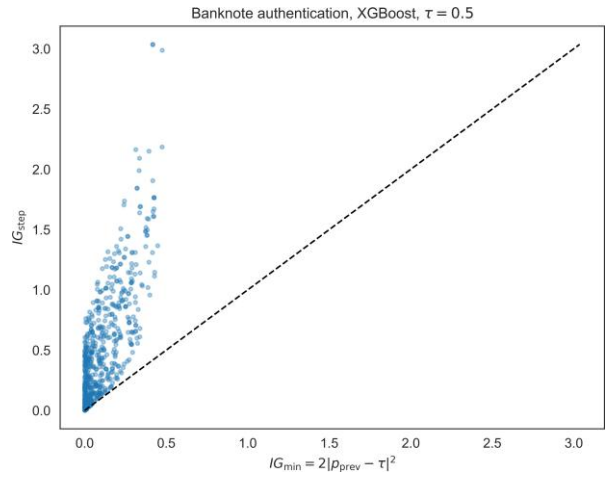
Figure S13: Threshold-certificate scatter diagnostics for the banknote-authentication dataset at $\tau = 0.3$.



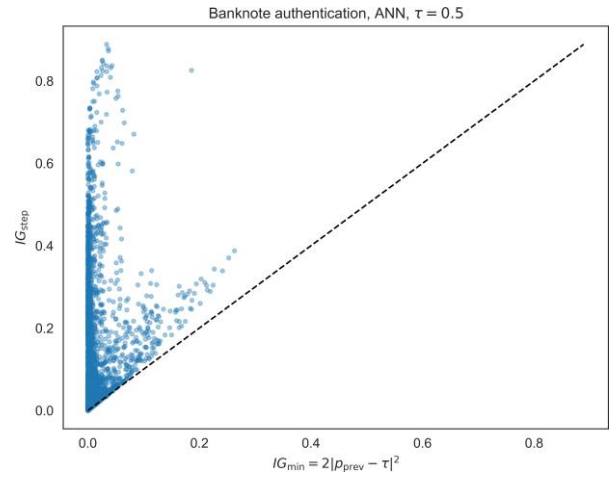
(a) Logistic regression



(b) Random forest

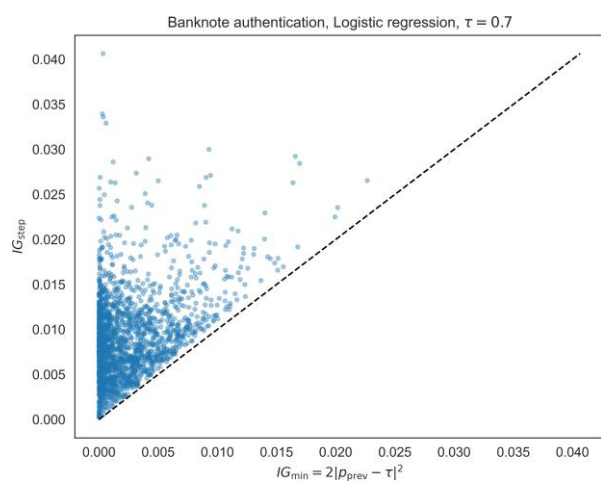


(c) XGBoost

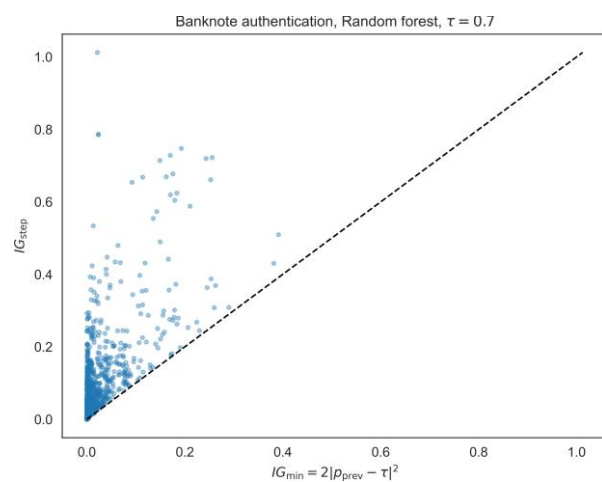


(d) ANN

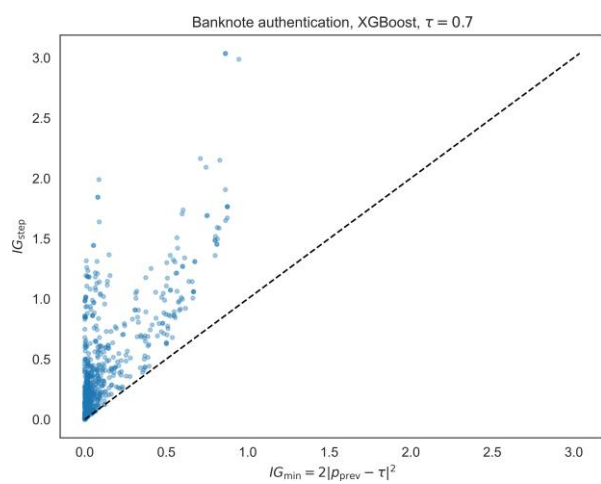
Figure S14: Threshold-certificate scatter diagnostics for the banknote-authentication dataset at $\tau = 0.5$.



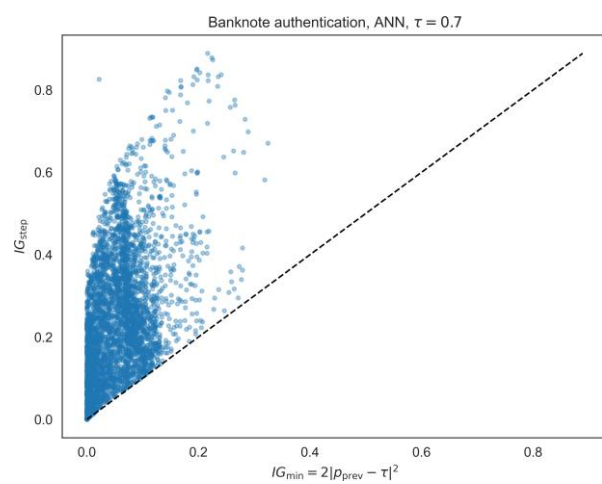
(a) Logistic regression



(b) Random forest



(c) XGBoost

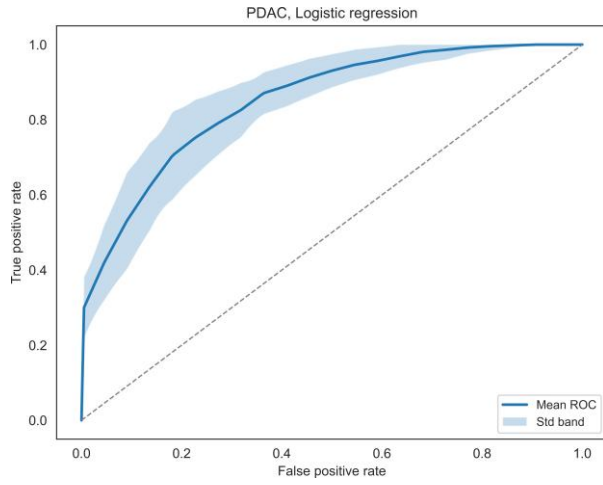


(d) ANN

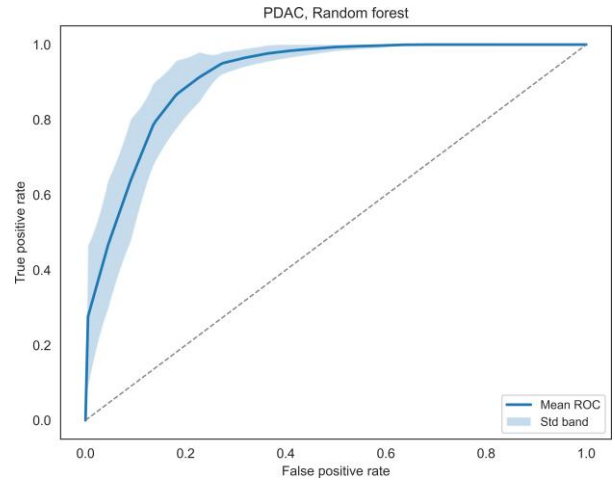
Figure S15: Threshold-certificate scatter diagnostics for the banknote-authentication dataset at $\tau = 0.7$.

2 Final ROC curves

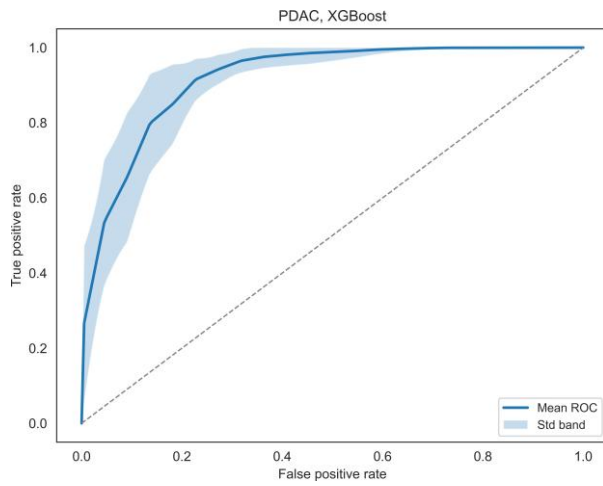
The following figures show final ROC curves for each dataset-model pair. Each curve is generated from the final progressive training size for the corresponding dataset. The shaded band is the standard-deviation band across the 20 repeated-holdout repetitions.



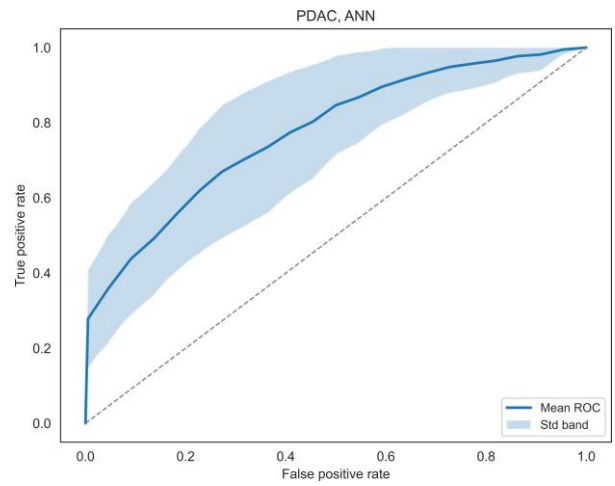
(a) Logistic regression



(b) Random forest

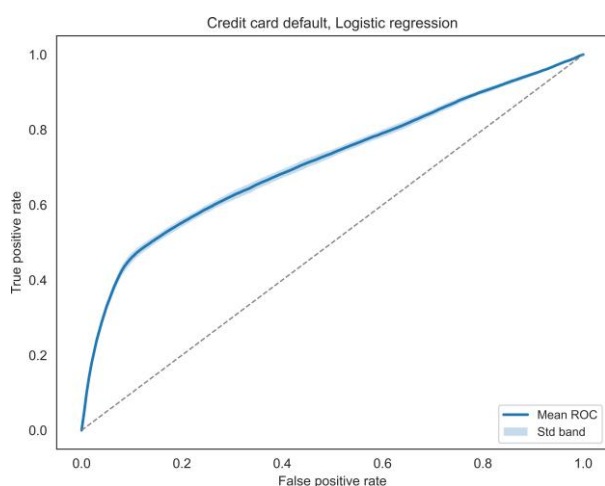


(c) XGBoost

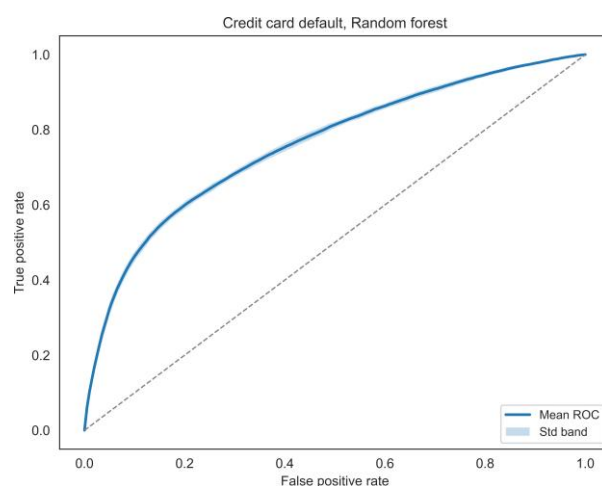


(d) ANN

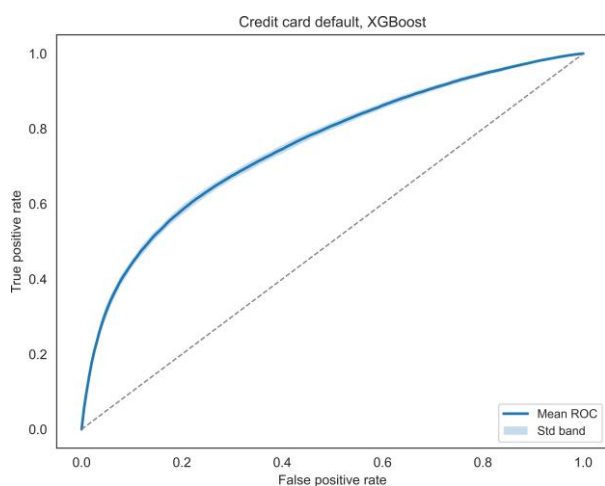
Figure S16: Final ROC curves for the PDAC biomarker dataset.



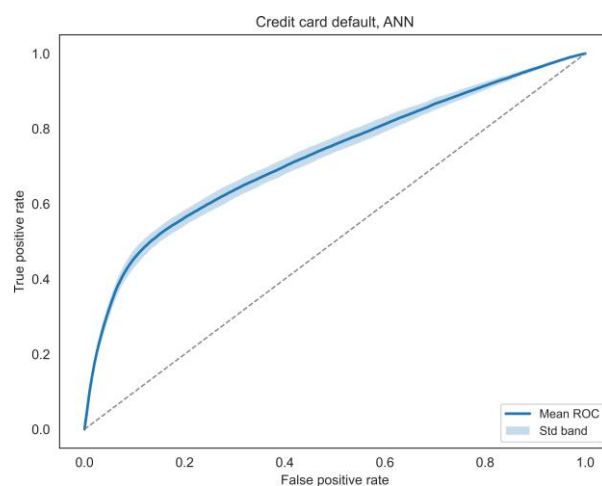
(a) Logistic regression



(b) Random forest

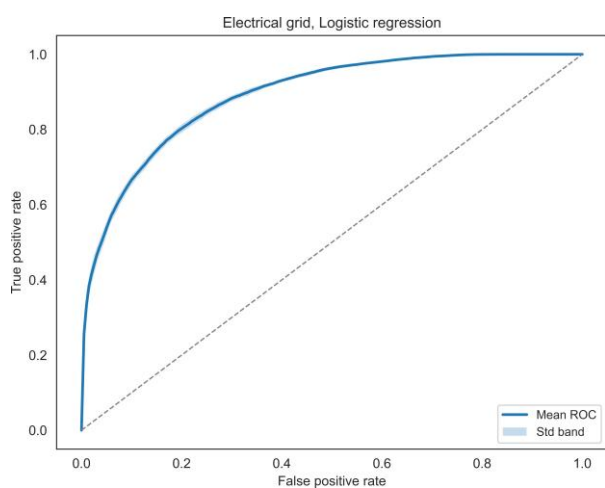


(c) XGBoost

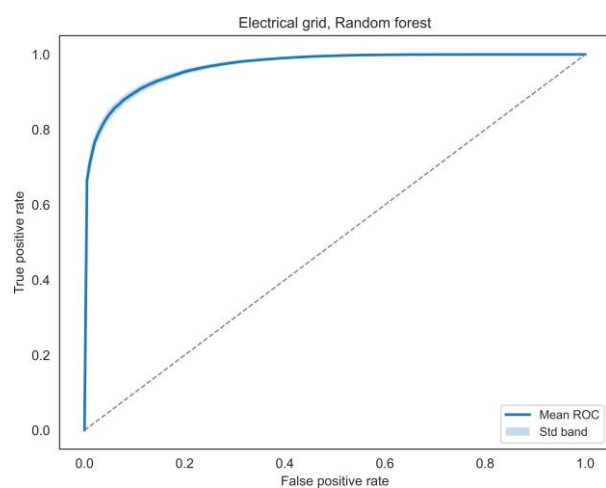


(d) ANN

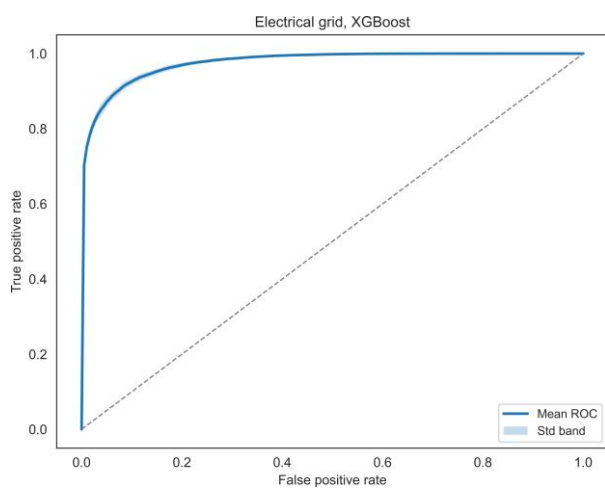
Figure S17: Final ROC curves for the credit-card-default dataset.



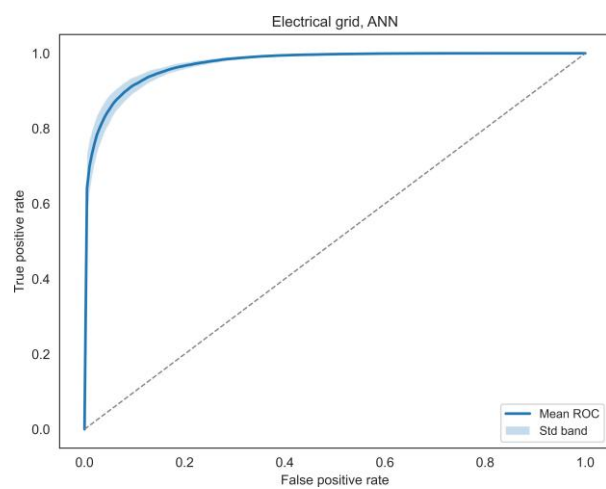
(a) Logistic regression



(b) Random forest

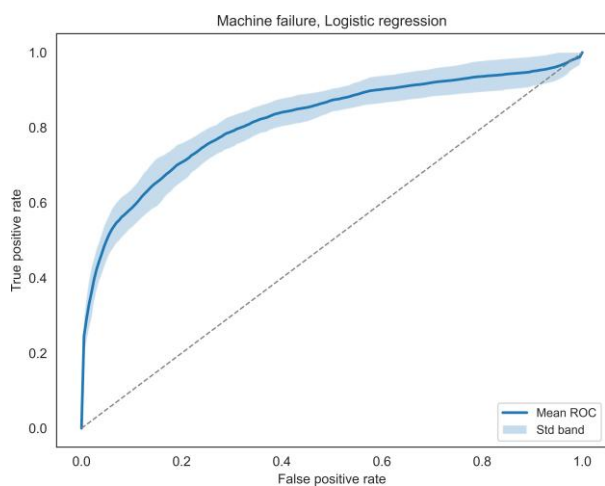


(c) XGBoost

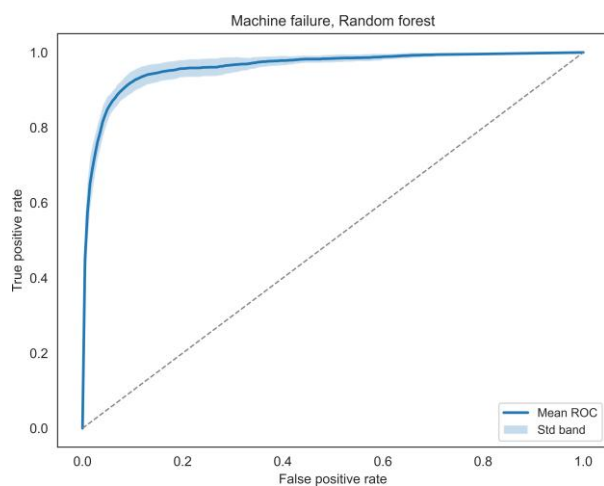


(d) ANN

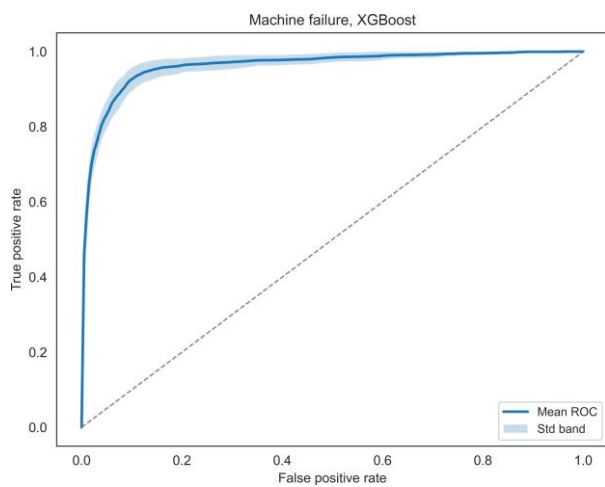
Figure S18: Final ROC curves for the electrical-grid-stability dataset.



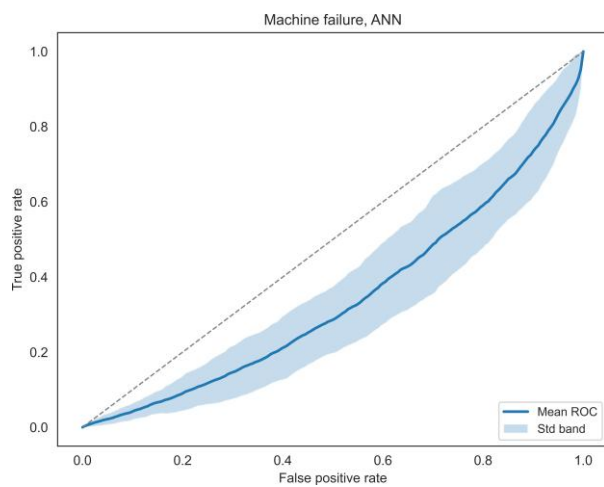
(a) Logistic regression



(b) Random forest

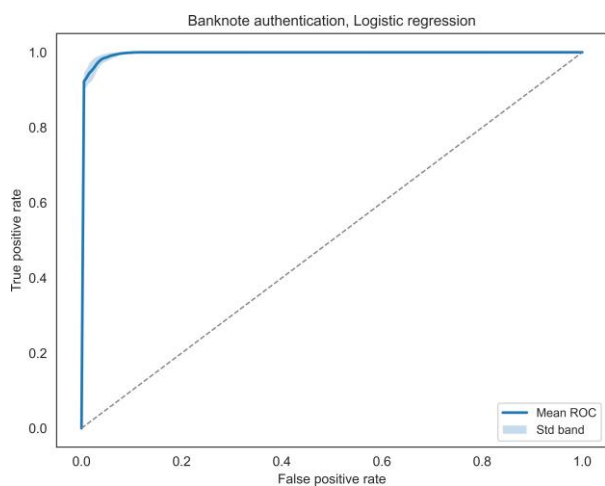


(c) XGBoost

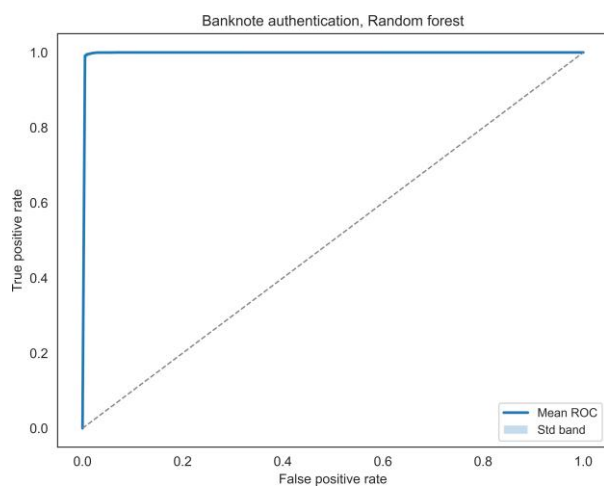


(d) ANN

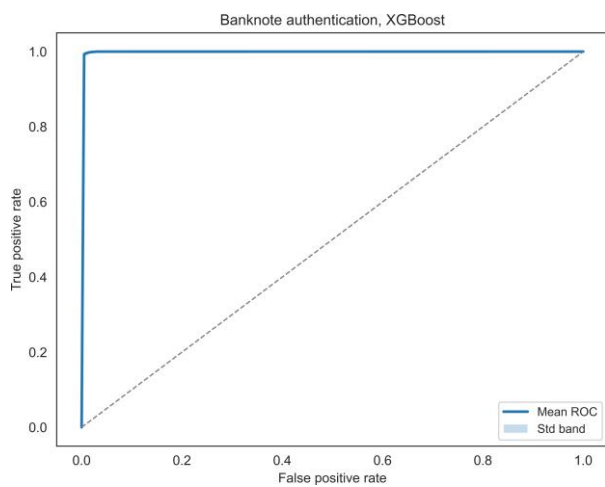
Figure S19: Final ROC curves for the machine-failure dataset.



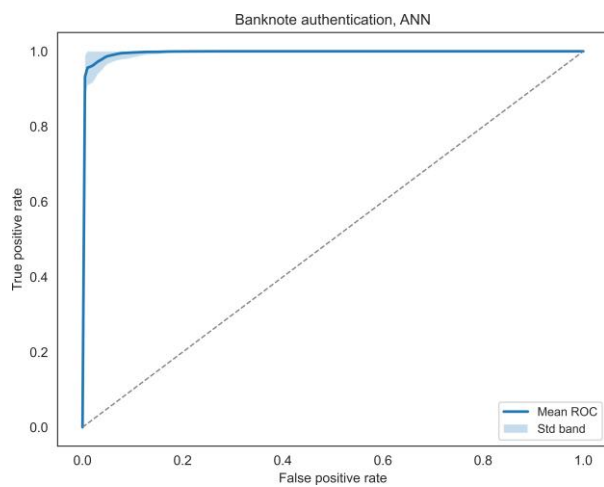
(a) Logistic regression



(b) Random forest



(c) XGBoost



(d) ANN

Figure S20: Final ROC curves for the banknote-authentication dataset.